Reverse-check review package — `DMsimp_gen`

An independent, **blank-slate** agent instance reconstructed the physics below from the sanitized `.fr` alone (no paper, no metadata, no conversation history). A second fresh instance then compared the reconstruction against the source paper. The final verdict belongs to the human reviewer — sign off at the bottom.

item	value
model file	`DMsimp/model/DMsimp_gen.fr`
original model name	`DMsimp_gen` (hidden from the agent)
paper	DMsimp/text/1508.00564.txt
blindness scope	blank slate w.r.t. paper, authors, model name, comments and prose labels — NOT w.r.t. field/parameter symbol names, which are kept so the reviewer can map the reconstruction back

Verbatim Lagrangian terms (from the `.fr`)

These are the terms the reconstruction must account for, quoted unmodified. Check each against its reconstructed LaTeX form below.

`L0X` (:=)

```
1/2 MXr gSXr Xr Xr Y0 + MXc gSXc Xcbar Xc Y0 + Xdbar.(gSXd + I gPXd Ga[5]).Xd Y0
```

`L0SM` (:=)

```
1/Sqrt[2] yt tbar.(gSt + I gPt Ga[5]).t Y0
```

`L0SMg` (:=)

```
1/Lambda FS[G,mu,nu,a] (gSg FS[G,mu,nu,a] + gPg Dual[FS][G,mu,nu,a]) Y0
```

`L0DM` (:=)

```
L0X + L0SM + L0SMg
```

`L1X` (:=)

```
I gVXc/2 (Xcbar del[Xc,mu] - del[Xcbar,mu] Xc) Y1[mu] + Xdbar.Ga[mu].(gVXd + gAXd Ga[5]).Xd
↪  Y1[mu]
```

`L1SM` (:=)

```
tbar.Ga[mu].(gVt + gAt Ga[5]).t Y1[mu] + bbar.Ga[mu].(-gAt Ga[5]).b Y1[mu]
```

`L1DM` (:=)

```
L1X + L1SM
```

Blank-slate reconstruction

Reconstructed Physics From `sanitized.fr`

Lagrangian

Conventions: repeated Lorentz and color indices are summed. The gluon field strength is

$$G^a_{\mu\nu} = \partial_\mu G^a_\nu - \partial_\nu G^a_\mu + g_s f^{abc} G^b_\mu G^c_\nu, \qquad \widetilde{G}^{a\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma} G^a_{\rho\sigma}.$$

No `DC[...]` covariant derivatives appear in the file. The only derivative acting on a new field is an ordinary partial derivative. All new fields have `Q -> 0` and no declared color or weak indices, so they are SM gauge singlets; their gauge covariant derivative would therefore reduce to

$$D_\mu = \partial_\mu$$

on the new fields. The SM quark and gluon objects use the standard FeynRules SM gauge content.

`L0X`

$$\mathcal{L}_{\texttt{L0X},\,X_r} = \frac{1}{2}\, M_{X_r}\, g_{SX_r}\, X_r X_r\, Y_0.$$

$$\mathcal{L}_{\texttt{L0X},\,X_c} = M_{X_c}\, g_{SX_c}\, X_c^\dagger X_c\, Y_0.$$

$$\mathcal{L}_{\texttt{L0X},\,X_d} = \bar{X}_d\, (g^S_{\rm DM} + i g^P_{\rm DM}\gamma^5)\, X_d\, Y_0.$$

`L0SM`

$$\mathcal{L}_{\texttt{L0SM}} = \frac{y_t}{\sqrt{2}}\, \bar{t}_i\, (g^S_t + i g^P_t \gamma^5)\, t_i\, Y_0.$$

Here i is the color index, contracted as δ_{ij}.

`L0SMg`

$$\mathcal{L}_{\texttt{L0SMg},\,S} = \frac{g^S_g}{\Lambda}\, Y_0\, G^a_{\mu\nu} G^{a\mu\nu}.$$

$$\mathcal{L}_{\texttt{L0SMg},\,P} = \frac{g^P_g}{\Lambda}\, Y_0\, G^a_{\mu\nu} \widetilde{G}^{a\mu\nu}.$$

`L0DM`

$$\mathcal{L}_{\texttt{L0DM}} = \mathcal{L}_{\texttt{L0X}} + \mathcal{L}_{\texttt{L0SM}} + \mathcal{L}_{\texttt{L0SMg}}.$$

`L1X`

$$\mathcal{L}_{\texttt{L1X},\,X_c} = \frac{i g_{VX_c}}{2}\, \left[X_c^\dagger \partial_\mu X_c - (\partial_\mu X_c^\dagger) X_c\right] Y_1^\mu.$$

$$\mathcal{L}_{\texttt{L1X},\,X_d} = \bar{X}_d \gamma^\mu\, (g^V_{\rm DM} + g^A_{\rm DM}\gamma^5)\, X_d\, Y_{1\mu}.$$

Equivalently, in chiral form,

$$\bar{X}_d \gamma^\mu \left[(g^V_{\rm DM} - g^A_{\rm DM}) P_L + (g^V_{\rm DM} + g^A_{\rm DM}) P_R\right] X_d\, Y_{1\mu}.$$

`L1SM`

$$\mathcal{L}_{\texttt{L1SM},\,t} = \bar{t}_i \gamma^\mu\, (g^V_t + g^A_t \gamma^5)\, t_i\, Y_{1\mu}.$$

Equivalently,

$$\bar{t}_i \gamma^\mu \left[(g^V_t - g^A_t) P_L + (g^V_t + g^A_t) P_R\right] t_i\, Y_{1\mu}.$$

$$\mathcal{L}_{\texttt{L1SM},\,b} = -\, g^A_t\, \bar{b}_i \gamma^\mu \gamma^5 b_i\, Y_{1\mu}.$$

Equivalently,

$$\mathcal{L}_{\texttt{L1SM},\,b} = \bar{b}_i \gamma^\mu \left[g^A_t P_L - g^A_t P_R\right] b_i\, Y_{1\mu}.$$

`L1DM`

$$\mathcal{L}_{\texttt{L1DM}} = \mathcal{L}_{\texttt{L1X}} + \mathcal{L}_{\texttt{L1SM}}.$$

Field Table

Symbol	Class	Spin	SU(3) rep	SU(2) rep	U(1) charge / hypercharge	Self-conjugate	Mass
X_r	`S[7]`	0	**1**	**1**	$Q = 0$, hence $Y = 0$ for an EW singlet	Yes, real scalar	$M_{X_r} = 10$
X_c	`S[8]`	0	**1**	**1**	$Q = 0$, hence $Y = 0$ for an EW singlet	No, complex scalar	$M_{X_c} = 10$
X_d	`F[7]`	1/2	**1**	**1**	$Q = 0$, hence $Y = 0$ for an EW singlet	No, Dirac fermion	$M_{X_d} = 10$
Y_0	`S[9]`	0	**1**	**1**	$Q = 0$, hence $Y = 0$ for an EW singlet	Yes, real scalar	$M_{Y_0} = 1000$
Y_1	`V[7]`	1	**1**	**1**	$Q = 0$, hence $Y = 0$ for an EW singlet	Yes, real vector	$M_{Y_1} = 1000$

Widths declared in the file are $\Gamma_{X_r} = 0$, $\Gamma_{X_c} = 0$, $\Gamma_{X_d} = 0$, $\Gamma_{Y_0} = W_{Y_0} = 10$, and $\Gamma_{Y_1} = W_{Y_1} = 10$.

Parameters

Symbol	Default value	Appears in	Physical meaning
`gSXr`	0	$\frac{1}{2} M_{X_r} g_{SX_r} X_r^2 Y_0$	Scalar coupling of the real scalar X_r to the scalar mediator Y_0.
`gSXc`	0	$M_{X_c} g_{SX_c} X_c^\dagger X_c Y_0$	Scalar coupling of the complex scalar X_c to Y_0.
`gSXd`	1	$\bar{X}_d g^S_{\mathrm{DM}} X_d Y_0$	Scalar Yukawa-like coupling of Dirac fermion X_d to Y_0.
`gPXd`	0	$i\bar{X}_d g^P_{\mathrm{DM}} \gamma^5 X_d Y_0$	Pseudoscalar coupling of X_d to Y_0.
`gSt`	1	$\frac{y_t}{\sqrt{2}} \bar{t} g^S_t t Y_0$	Scalar coupling of Y_0 to top quarks, normalized by $y_t/\sqrt{2}$.
`gPt`	0	$\frac{y_t}{\sqrt{2}} i\bar{t} g^P_t \gamma^5 t Y_0$	Pseudoscalar coupling of Y_0 to top quarks, normalized by $y_t/\sqrt{2}$.

Symbol	Default value	Appears in	Physical meaning
`Lambda`	10000	$\Lambda^{-1} Y_0 G_{\mu\nu} G^{\mu\nu}$, $\Lambda^{-1} Y_0 G_{\mu\nu} \widetilde{G}^{\mu\nu}$	Heavy scale suppressing the effective scalar and pseudoscalar gluon operators.
`gSg`	0	$\frac{g_g^S}{\Lambda} Y_0 G^a_{\mu\nu} G^{a\mu\nu}$	Effective CP-even coupling of Y_0 to gluons.
`gPg`	0	$\frac{g_g^P}{\Lambda} Y_0 G^a_{\mu\nu} \widetilde{G}^{a\mu\nu}$	Effective CP-odd coupling of Y_0 to gluons.
`gVXc`	0	$\frac{i g_{VX_c}}{2} X_c^\dagger \overleftrightarrow{\partial_\mu} X_c Y_1^\mu$	Vector-current coupling of complex scalar X_c to vector mediator Y_1.
`gVXd`	1	$\bar{X}_d \gamma^\mu g^V_{\mathrm{DM}} X_d Y_{1\mu}$	Vector coupling of Dirac fermion X_d to Y_1.
`gAXd`	0	$\bar{X}_d \gamma^\mu g^A_{\mathrm{DM}} \gamma^5 X_d Y_{1\mu}$	Axial-vector coupling of X_d to Y_1.
`gVt`	1	$\bar{t} \gamma^\mu g^V_t t Y_{1\mu}$	Vector coupling of Y_1 to top quarks.
`gAt`	0	$\bar{t} \gamma^\mu g^A_t \gamma^5 t Y_{1\mu}$, $-\bar{b} \gamma^\mu g^A_t \gamma^5 b Y_{1\mu}$	Axial-vector coupling of Y_1 to tops and, with opposite sign structure in the Lagrangian, to bottoms.

Physics Summary

The file encodes a simplified SM-singlet dark-sector mediator setup with three possible dark matter candidates: a real scalar X_r, a complex scalar X_c, and a Dirac fermion X_d. A scalar mediator Y_0 couples to these dark states, to top quarks through scalar and pseudoscalar bilinears, and to gluons through dimension-five CP-even and CP-odd operators; a vector mediator Y_1 couples to scalar and fermion dark currents, to top vector/axial currents, and to a bottom axial current.

The interactions mediate production or annihilation channels such as $gg \to Y_0 \to X\bar{X}$, $t\bar{t} \to Y_0/Y_1 \to X\bar{X}$, mediator-associated heavy-quark processes, and mediator decays into dark matter, tops, bottoms, or gluons depending on the enabled couplings and masses.

Paper cross-check

Comparison of `reconstruction.md` Against the Paper Model

Paper Model Location

The paper defines the simplified model in **Section 2, "Simplified Model"**. It states that the mediator couples only to top quarks and dark matter, with a bottom-quark coupling added only for the axial-vector case to cancel the gauge anomaly. The dark matter particle χ is taken to be a **Dirac fermion** in the study, although the implementation is described as flexible enough to allow real or complex scalar dark matter.

The interaction Lagrangians are:

- **Eq. (1)**: scalar mediator Y_0

$$\mathcal{L}^{Y_0}_{DM} = \bar{\chi}(g^S_{DM} + i g^P_{DM} \gamma^5)\chi Y_0, \qquad \mathcal{L}^{Y_0}_{SM} = \bar{t}\,\frac{y_t}{\sqrt{2}}(g^S_t + i g^P_t \gamma^5) t Y_0.$$

- **Eq. (2)**: vector mediator Y_1

$$\mathcal{L}_{DM}^{Y_1} = \bar{\chi}\gamma^\mu (g_{DM}^V + g_{DM}^A \gamma^5)\chi Y_{1\mu},$$
$$\mathcal{L}_{SM}^{Y_1} = \bar{t}\gamma^\mu (g_t^V + g_t^A \gamma^5) t Y_{1\mu} + \bar{b}\gamma^\mu (-g_t^A \gamma^5) b Y_{1\mu}.$$

- **Eq. (3)** appears later in Section 4.2 as an **infinite-top-mass EFT approximation**, not as the primary simplified-model Lagrangian:

$$\mathcal{L} = \frac{\alpha_s}{12\pi v} g_t^S \, G_{\mu\nu} G^{\mu\nu} Y_0 + \frac{\alpha_s}{8\pi v} g_t^P \, G_{\mu\nu} \tilde{G}^{\mu\nu} Y_0.$$

Term-by-Term Comparison

paper term (eq. ref)	reconstruction term	verdict	notes
Dirac dark matter χ, Section 2	X_d, Dirac fermion, SM singlet	agree	The paper uses χ; the reconstruction uses X_d. Physics content matches for the fermionic DM used in the study.
Scalar mediator Y_0, Section 2 / Eq. (1)	Y_0, real scalar SM singlet	agree	The paper does not spell out gauge reps, but the mediator is a neutral s-channel boson; the reconstruction's singlet assignment is consistent.
Vector mediator Y_1, Section 2 / Eq. (2)	Y_1, real vector SM singlet	agree	Consistent with the paper's neutral vector mediator.
$\bar{\chi}(g_{DM}^S + i g_{DM}^P \gamma^5)\chi Y_0$, Eq. (1)	$\bar{X}_d(g_{DM}^S + i g_{DM}^P \gamma^5) X_d Y_0$	agree	Same scalar and pseudoscalar Dirac bilinears, same $i\gamma^5$ structure.
$\bar{t}\, \frac{y_t}{\sqrt{2}}(g_t^S + i g_t^P \gamma^5) t Y_0$, Eq. (1)	$\frac{y_t}{\sqrt{2}}\bar{t}_i(g_t^S + i g_t^P \gamma^5) t_i Y_0$	agree	Same top coupling and Yukawa normalization. Reconstruction makes the color contraction explicit.
$\bar{\chi}\gamma^\mu(g_{DM}^V + g_{DM}^A \gamma^5)\chi Y_{1\mu}$, Eq. (2)	$\bar{X}_d\gamma^\mu(g_{DM}^V + g_{DM}^A \gamma^5) X_d Y_{1\mu}$	agree	Same vector and axial-vector Dirac-current structure.
$\bar{t}\gamma^\mu(g_t^V + g_t^A \gamma^5) t Y_{1\mu}$, Eq. (2)	$\bar{t}_i\gamma^\mu(g_t^V + g_t^A \gamma^5) t_i Y_{1\mu}$	agree	Same top vector and axial couplings; reconstruction makes color explicit.
$\bar{b}\gamma^\mu(-g_t^A \gamma^5) b Y_{1\mu}$, Eq. (2)	$-g_t^A \bar{b}_i \gamma^\mu \gamma^5 b_i Y_{1\mu}$	agree	Same bottom axial coupling, with opposite sign to the top axial coupling as required for anomaly cancellation.
Bottom coupling introduced only for axial-vector anomaly cancellation, Section 2 after Eq. (2)	Bottom term tied only to g_t^A, no bottom vector term	agree	Reconstruction correctly omits a bottom vector coupling and includes only the axial structure.

paper term (eq. ref)	reconstruction term	verdict	notes
Top-EFT scalar gluon operator $\frac{\alpha_s}{12\pi v} g_t^S G_{\mu\nu} G^{\mu\nu} Y_0$, Eq. (3)	$\frac{g_g^S}{\Lambda} Y_0 G^a_{\mu\nu} G^{a\mu\nu}$	disagree	Same operator shape, but different coefficient structure: paper's Eq. (3) is fixed by $\alpha_s/(12\pi v)\, g_t^S$, while reconstruction has an independent g_g^S/Λ.
Top-EFT pseudoscalar gluon operator $\frac{\alpha_s}{8\pi v} g_t^P G_{\mu\nu} \tilde{G}^{\mu\nu} Y_0$, Eq. (3)	$\frac{g_g^P}{\Lambda} Y_0 G^a_{\mu\nu} \widetilde{G}^{a\mu\nu}$	disagree	Same CP-odd operator shape, but paper's coefficient is $\alpha_s/(8\pi v)\, g_t^P$; reconstruction uses independent g_g^P/Λ.
Base simplified model in Section 2 has no independent $Y_0 GG$ or $Y_0 G\tilde{G}$ contact term	`LOSMg` included in `LODM`	extra-in-reconstruction	Eq. (3) is discussed as an infinite-top-mass approximation later, not as an independent base-model interaction with free g_g/Λ coefficients.
Paper study takes χ to be a Dirac fermion, Section 2	X_r real scalar DM with $\frac{1}{2} M_{X_r} g_{SX_r} X_r^2 Y_0$	extra-in-reconstruction	The paper says the implementation is flexible to allow scalar DM, but the displayed paper Lagrangian and study use Dirac χ.
Paper study takes χ to be a Dirac fermion, Section 2	X_c complex scalar DM with $M_{X_c} g_{SX_c} X_c^\dagger X_c Y_0$	extra-in-reconstruction	Same caveat: plausible implementation extension, but not part of the explicit paper Lagrangian.
Paper Eq. (2) gives vector mediator coupling to fermionic DM only	$X_c^\dagger \overleftrightarrow{\partial_\mu} X_c Y_1^\mu$	extra-in-reconstruction	The complex-scalar vector current is not present in the paper's displayed Lagrangian.
Masses and widths are free/benchmark-dependent, Section 2 and Tables 1-2	Fixed defaults $M_X = 10$, $M_Y = 1000$, $\Gamma_Y = 10$	disagree	The paper treats masses and widths as parameters and gives benchmark values; reconstruction reports implementation defaults that do not correspond to the paper's benchmark table.
Scalar real-DM coupling in paper	None explicitly displayed	missing-in-reconstruction	Not applicable as a paper term: the paper does not provide explicit real-scalar DM Lagrangian terms, so there is no paper-defined scalar-DM term missing from the reconstruction.

paper term (eq. ref)	reconstruction term	verdict	notes
Complex scalar-DM coupling in paper	None explicitly displayed	missing-in-reconstruction	Not applicable as a paper term: scalar DM is mentioned as implementation flexibility, not defined term-by-term in the paper.

Disagreements and Checks

- **Independent gluon contact operators** — severity: **substantive**. A human should check whether the implementation intentionally extends the paper model with free g_g^S/Λ and g_g^P/Λ operators, or whether these were meant to reproduce Eq. (3), in which case the coefficient mapping is missing.
- **Eq. (3) coefficient mismatch** — severity: **substantive**. A human should check whether g_g^S/Λ and g_g^P/Λ are externally constrained to $\alpha_s g_t^S/(12\pi v)$ and $\alpha_s g_t^P/(8\pi v)$; without that mapping, the reconstruction describes a more general EFT than the paper's top-EFT approximation.
- **Real scalar dark matter X_r** — severity: **convention**. A human should check whether the review target is the paper's displayed study model, which uses Dirac χ, or the broader implementation alluded to by the paper as allowing real scalar dark matter.
- **Complex scalar dark matter X_c** — severity: **convention**. A human should check whether the implementation file was expected to include scalar-DM options beyond the paper's explicit Lagrangian.
- **Complex scalar vector-current coupling to Y_1** — severity: **convention**. A human should check whether this is an implementation extension for scalar DM rather than a mismatch with the fermionic paper model.
- **Default masses and widths** — severity: **cosmetic**. A human should check whether the reconstruction is reporting UFO/FeynRules default parameter values, because the paper's physical benchmarks use different mass and width choices.

Overall Assessment

The reconstruction matches the paper's explicit fermionic simplified-model Lagrangian in Eq. (1) and Eq. (2): the scalar/pseudoscalar Y_0 couplings to Dirac dark matter and tops, the vector/axial Y_1 couplings to Dirac dark matter and tops, and the special bottom axial coupling all have the correct field content, chirality, signs, and normalization. The main differences are that the reconstruction contains implementation-level extensions not present in the displayed paper model, especially real and complex scalar dark matter interactions and independent $Y_0 GG$, $Y_0 G\tilde{G}$ operators. The gluon operators are physically close in form to the paper's Eq. (3), but their coefficient structure is not the same unless an external parameter mapping is imposed.

Suggested checks for the reviewer

1. Every verbatim `.fr` term above has a reconstructed LaTeX counterpart with the same field content, chirality and conjugation.
2. Kinetic terms: covariant derivative gauge content matches the field's representations; normalization is canonical.
3. Non-self-conjugate interaction terms appear together with their Hermitian conjugates.
4. Quantum numbers in the field table match the `.fr` declarations (and the paper, where the cross-check table flags disagreements).
5. Numeric masses/couplings are placeholders unless the paper pins them — treat values as demo inputs, not measurements.
6. Sanitizer scope: 19 prose labels were scrubbed; field/parameter symbol names were kept and may hint at the model's identity.

Physicist sign-off

- Reviewed by: ________________________________ Date: __________________
- Verdict: [ ] approve [ ] approve with corrections [ ] reject
- Notes: